

Exact constant-field Landau–Lifshitz–Gilbert flow on the sphere

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Abstract

We close the complete source-local dynamics of the constant-field Landau–Lifshitz–Gilbert equation on the unit sphere. In the stereographic coordinate $z = (m_1 + im_2)/(1 + m_3)$, the nonlinear vector field becomes $\dot{z} = (-\alpha\omega + i\omega)z$, while m_3 is an exact tanh trajectory. The energy law, north/south stability, pure-precession latitude periods, identity face, and every sampled-time fixed set follow in one boundary atlas. The periodic face is a continuum and therefore does not provide an isolated primitive-orbit or arithmetic determinant. Independent exact reconstruction and symbolic checks accompany the paper.

1 Model and exact chart

Let $e_3 = (0, 0, 1)$, $m(t) \in \mathbb{S}^2$, and $\alpha, \omega \geq 0$. We freeze

$$\dot{m} = -\omega m \times e_3 - \alpha\omega m \times (m \times e_3). \quad (1)$$

This is the normalized Landau–Lifshitz/Gilbert convention: the usual factor $1 + \alpha^2$ in the unreduced Gilbert equation has been absorbed into the declared frequency ω (equivalently, into the time unit). The vector field is tangent to the sphere. On $m_3 > -1$ put $z = (m_1 + im_2)/(1 + m_3)$. The inverse formulas are

$$m_1 + im_2 = \frac{2z}{1 + |z|^2}, \quad m_3 = \frac{1 - |z|^2}{1 + |z|^2}. \quad (2)$$

Theorem 1 (Exact flow and pole atlas). *For $-1 < m_3(0) < 1$, equation (1) has*

$$z(t) = z(0)e^{(-\alpha\omega + i\omega)t}, \quad m_3(t) = \tanh(\alpha\omega t + \operatorname{artanh} m_3(0)). \quad (3)$$

The two poles are equilibria. With $E = 1 - m_3$,

$$\dot{E} = -\alpha\omega(1 - m_3^2) \leq 0. \quad (4)$$

If $\alpha\omega > 0$, the north pole is asymptotically stable and the south pole is unstable. Their transverse eigenvalues have real parts $-\alpha\omega$ and $+\alpha\omega$, respectively, and precession frequency ω .

Proof. Writing (1) in components gives $\dot{m}_1 = -\omega m_2 - \alpha\omega m_1 m_3$, $\dot{m}_2 = \omega m_1 - \alpha\omega m_2 m_3$, and $\dot{m}_3 = \alpha\omega(m_1^2 + m_2^2)$. The complex combination and the unit-sphere identity yield the scalar linear equation and the logistic equation in (3). Substitution in (2) proves reconstruction and (4). Linearizing the complex equation at the two chart ends gives the stated real parts; the pole cases themselves are immediate from (1). $\square$

2 Periodic and sampled-time faces

Proposition 1 (All boundary cases). *If $\alpha = 0$ and $\omega > 0$, latitude is conserved and every nonpolar point runs on a circle of period $2\pi/\omega$. If $\omega = 0$, the flow is the identity for every α . For a sampled time $\tau > 0$, when $\alpha\omega > 0$ the fixed set is exactly the two poles. On the pure precession face it is all of $\mathbb{S}^2$ iff $\omega\tau \in 2\pi\mathbb{Z}$, and otherwise it is the two poles; the $\omega = 0$ and $\tau = 0$ faces are identity maps.*

Proof. Equation (3) preserves $|z|$ exactly when $\alpha = 0$, and its phase is ωt . If $\alpha > 0$, the modulus can return to its initial value only at a pole. The listed resonance and identity alternatives then follow directly from the exponential flow. $\square$

References

- [1] M. Lakshmanan, “The fascinating world of the Landau–Lifshitz–Gilbert equation: an overview,” *Philosophical Transactions of the Royal Society A* 369(1935), 1280–1300 (2011). DOI.

Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; the source theorem contains no target arithmetic, Euler-factor, target-zero, automorphy or Hilbert–Pólya claim. **Data and code.** Exact formulas, regression rows and independent audit programs accompany HCS-C234. **AI-use disclosure.** Generative tools assisted drafting and code generation; internal checks validate the displayed identities. This is not external peer review.